



Application Note: AS3935-AN05 – Automatic Antenna Tuning

AS3935

Automatic Antenna Tuning

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Revision History

Revision	Date	Owner	Description
1.0	06.06.2012	JPE	Initial release
1.1	20.02.2013	JRY	Update to corporate format

1 General Description

This application explains the implementation of the automatic antenna tuning algorithm.

2 Antenna Tuning

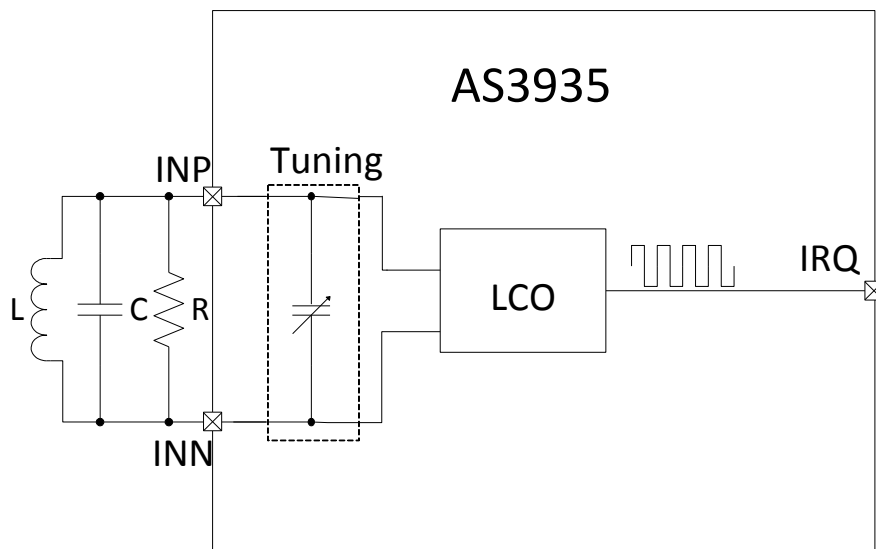
The antenna tuning algorithm allows tuning the antenna exactly to 500 kHz with the help of the internal capacitance array. Tuning the antenna is required because of the following items:

- Check if antenna is correctly assembled to exclude soldering issues.
- The calibration of the internal RCOs (TRCO and SRCO) is derived from the resonance frequency of the antenna. That means the precision of the RCOs depends on the precision of the LC-tank of the antenna.
- The statistical analysis is based on a tuned 500 kHz antenna with a Q-factor of 15.

The antenna tuning algorithm can be performed in the following way:

1. Measure actual resonance frequency of the antenna on the IRQ pin.
2. Add additional internal capacitance in parallel to the external LC.
3. Repeat this sequence until the resonance frequency is tuned to 500 kHz.

Figure 1: Tuning Capacitor Block Diagram



2.1 Measure LCO Frequency

Connect the LCO to the IRQ pin setting register REG0x08[7] = 1. Select the division ratio of the LCO frequency defined in register REG0x03[7:6]. Based on the division ratio the theoretical period can be calculated. Wait 2ms for the LCO to start up and to be settled before measuring the frequency.

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